

Juan Arturo Abaurrea Calafell

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Education

Bielefeld University *Oct 2026 – Sep 2029, Bielefeld, Germany*
PhD in Intelligent Systems at CITEC. MSCA Doctoral Network Fellow (Spring DC5).

Polytechnic University of Madrid (UPM) *Sep 2025 – Jul 2026, Madrid, Spain*
Master's Degree in Machine Learning and Big Data. Class representative.

- **Grade:** 8.78 out of 10.
- **Relevant Courses:** Management of Big Data Systems (9.8), Deep Learning for Natural Language Processing (10), DL for Computer Vision (7.5), Reinforcement Learning and Generative Techniques (8.6), AI Models for Time Series (9.7), Social Network Analysis (8.7), Data Security and Privacy (8.4).

University of the Balearic Islands (UIB) *Sep 2021 – Jun 2025, Palma de Mallorca, Spain*
Bachelor's Degree in Computer Science and Engineering – Artificial Intelligence and Computing track.

- **Grade:** 8.3 out of 10 (8 honors, [top 5%](#))
- **Relevant Courses:** Machine Learning (9.2), Artificial Intelligence (honor), Intelligent Systems (9.0), Data Mining (7.0), Advanced Algorithms (honor), Databases I & II (7.0 & honor), Statistics (8.5).

Ulm University of Applied Sciences (THU) *Mar 2024 – Jul 2024, Ulm, Germany*
Exchange semester. Completed Bachelor's thesis and three additional subjects.

- **Grade:** 9.7 out of 10.
- **Bachelor's Thesis (9.6):** Implemented a chatbot for colonoscopy preparation using NLP techniques.

Summer/Winter Schools

- **XJTU** – IT Summer School (full scholarship) *Aug 2026 – Sep 2026, Xi'an, China*
- **ZHAW** – HealthTech Summer School (tuition waived) *Jul 2026, Zurich, Switzerland*
- **BIT** – AI Winter School (full scholarship) *Jan 2026, Zhuhai, China*

Experience

MTP, Entry Machine Learning Researcher *Feb 2026 – Jul 2026, Madrid, Spain*
Conducted Master's thesis research on backdoor detection in deep neural networks by analyzing internal activations and latent space representations across multiple layers ([see project below](#)).

TravelgateX, Entry Software Engineer *Oct 2024 – Dec 2024, Palma de Mallorca, Spain*
Backend development using C# and Visual Basic.

Languages

Native: Spanish, Catalan | **C1:** English (Cambridge certified) | **A1:** German

AI & Data-Related Skills

- **Programming Languages:** Python, R, SQL (MySQL, MariaDB, PostgreSQL), MATLAB
- **Machine Learning & AI:** PyTorch, Keras, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, torchvision, YOLO
- **Data Science:** NumPy, Pandas
- **Data:** Apache Spark (PySpark), Apache Kafka, RabbitMQ – Matplotlib, Seaborn, Plotly, Dash, Grafana
- **MLOps & Cloud Infrastructure:** Docker, Git, GitHub Actions, Azure, Jupyter Notebook

Honors & Awards

[High Academic Distinction – Top 5% of a 60-student cohort](#)

2025

- Officially recognized with a certificate for achieving one of the highest academic records in the Bachelor's Degree in Computer Science and Engineering (Artificial Intelligence and Computing).

[Cybersecurity AI Datathon – 2nd Place](#)

Nov 2025

- Achieved 0.884 F1-score on ransomware classification using Random Forest and LightGBM ensemble.
- Jailbroke custom LLM with prompt engineering techniques to extract passwords from it.

Relevant Projects

[Turning Neural Networks Against Each Other: Multi-Layer Backdoor Detection in Neural Networks](#)

Feb 2026 – Jun 2026

- Surveyed and taxonomized backdoor attacks/defences, identifying the single-layer assumption as the key limitation of existing activation-based detectors.
- Designed and evaluated four detectors (CAP, BMD, ZRF, Whistleblower) operating over whole-network activations rather than a single bottleneck layer.
- Showed backdoor information is distributed across layers; ZRF matched and exceeded multiple state-of-the-art methods, while the supervised Whistleblower detector generalized to unseen attacks.

[Backdoor Detection and Mitigation in Neural Networks](#)

Nov 2025 – Dec 2025

- Comprehensive survey of 70+ papers on neural network security, covering backdoor attacks, detection methods, and mitigation techniques.

[Spiking Neural Networks: A Comprehensive Literature Review](#)

Oct 2025 – Dec 2025

- Analyzed 50+ research papers on brain-inspired computing, covering neuron models, learning algorithms, and neuromorphic hardware.

[Multi-Author Writing Style Analysis](#)

Oct 2025 – Nov 2025

- Developed transformer-based models for detecting authorship changes for PAN@CLEF 2025 task.
- Achieved 92.2% F1 score using fine-tuned DeBERTa-v3-small.

[Interactive Earthquake Dashboard](#)

Oct 2025 – Nov 2025

- Built interactive Dash/Plotly dashboard visualizing earthquake data.

[Image Classification, Detection and Segmentation Using Neural Networks](#)

Dec 2024 – Jan 2025

- Implemented five deep learning models (ResNet50, VGG16, Inception v3, custom CNNs) achieving 90-96% F1-score with minimal training data.
- Developed object detection and segmentation pipelines using YOLOv8m and Mask R-CNN.

[Real Estate Price Analysis Using Data Mining](#)

Dec 2024 – Jan 2025

- Applied K-NN, Random Forest, and Linear Regression on 114K samples achieving <5% error for property predictions and clustering analysis.

[Chatbot for Colonoscopy Preparation](#)

Mar 2024 – Nov 2024

- Developed NLP-based chatbot using Levenshtein distance, TF-IDF, BM25, and sentence embeddings.

Other Technical Skills

- **Programming Languages:** Ada, Assembly, Bash, C, C#, C++, CSS, GoLang, HTML, Java, JavaScript, Lisp, PHP, Prolog, Visual Basic
- **Tools & Technologies:** Bootstrap, Docker, Fiddler, Git, GitHub, HubSpot, Insomnia, Jira, NetBeans, Postman, PyCharm, Visual Studio, Visual Studio Code, Wireshark, XAMPP